SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY
COMPUTER SCIENCE & INFORMATION TECHNOLOGY DEPARTMENT

Spring 2026
Machine Learning (CS-466)
Assignment # 1

Semester: 8th	Batch: 2022F
Announced Date: 14/04/2026	Due Date:24/04/2026
Total Marks: 03	
Instructor: Ms. Tahmina Khan	Obtained marks: ________

Submitted by: ____________________________ **Roll #:** ____________________ **Section:** _____

CLO #	Course Learning Outcomes (CLOs)	PLO Mapping	Bloom's Taxonomy
1	**Describe** the characteristics of different algorithms of Machine Learning and identify the functionalities of these in Classification and prediction.	PLO_2 (Knowledge for Solving Computing Problems)	C2 (Understanding)

Question No. 01: **[3 Marks]**

Discuss and **compare** the performance of following algorithms for classification:

(i) KNN
(ii) Naïve Bayes
(iii) Decision tree
(iv) Logistic regression

Download public dataset from Kaggle, GitHub or UCI Machine Learning repository. Write a code to evaluate performance of each algorithm using accuracy, precision, recall, F1- score metrics. Also plot confusion matrix and classification report of all algorithms.

On the basis of evaluation metrics, **select** the best model.

****Bonus:** Implement a simple ensemble method (e.g. bagging or boosting) using one of the above algorithms and evaluate its performance.

Note: You can use Python with scikit-learn or TensorFlow libraries to implement the algorithms and evaluate their performance.